
THE STANDARDIZATION OF THE COGNITIVE EMOTIONAL REGULATION QUESTIONNAIRE (CERQ) ON ROMANIAN POPULATION

Adela PERȚE^{*1}, Mircea MICLEA^{1, 2}

¹Cognitrom LTD, Cluj-Napoca, Romania

²Department of Psychology, Babes-Bolyai University, Cluj-Napoca, Romania

ABSTRACT

This study presents the Romanian version of the Cognitive Emotional Regulation Questionnaire (CERQ). Psychometric properties were assessed in different clinical and non-clinical groups of adolescents, adults and psychiatric patients (N = 1807) from different parts of the country. Results provide evidence for the reliability and validity of CERQ in relation to personality traits, symptoms of anxiety and depression, coping measures for the adolescent and adult groups, pathological conditions. The results show that CERQ is a useful instrument for assessing cognitive emotional coping strategies in the Romanian population.

KEYWORDS: *cognitive coping, emotion regulation strategies, validity, emotions, negative life events*

INTRODUCTION

Coping is a process that unfolds in the context of a situation or condition that is appraised as personally significant and as taxing or exceeding the individual's resources for coping (Lazarus & Folkman, 1984). It is a complex, multidimensional process that is sensitive both to environment, its demands and resources, and to personality dispositions.

Monat and Lazarus (1991, p. 5) offer a definition that refers to coping as the „individual's efforts to master demands (conditions of harm, threat, challenge) that are appraised (or perceived) as exceeding or taxing one's resources”. There is a classic distinction in the literature between *problem focused coping* (includes all coping strategies addressing directly the stressor), and *emotion focused coping* (refers to all coping strategies aimed at regulating the emotions associated with the

* Corresponding author:
E.mail: adela.perte@gmail.com

stressor) (Compas, Orosan, & Grant, 1993). Traditionally, problem focused coping strategies were considered more functional than emotion focused coping strategies (Thoits, 1995). However, the contextual approach to coping that guides much of coping research states explicitly that coping processes are not inherently good or bad (Lazarus & Folkman, 1984), rather the way that a coping strategy is being used, when or for how long, makes it more or less functional. A given coping strategy may be effective in one situation, but not in another, depending, for example, on the extent to which a situation is controllable (Folkman & Moskowitz, 2004). Hence, we cannot say that a certain coping strategy is a good one or a bad one. There are other factors that need to be considered when we evaluate coping strategies, such as the context, the time, the stressor. A coping strategy might be considered effective at the outset of a stressful situation, but may be ineffective later on (Folkman & Moskowitz, 2004). Coping processes are not independent processes; they are initiated in an emotional environment, so we must take into account all the factors when we evaluate them. Research in the field has shown that coping is strongly associated with the regulation of emotion, especially distress, throughout the stress process and that certain kinds of escapist coping strategies are consistently associated with poor mental health outcomes, while other kinds of coping are sometimes associated with negative outcomes, sometimes with positive ones (Folkman & Moskowitz, 2004). Emotion regulation is assumed to be an important factor in determining well being and/or successful functioning. (Cicchetti, Ackerman & Izard, 1995; Thompson, 1991). In the literature the concept of emotion regulation and coping are often used as interchangeable. Generally speaking, both concepts can be understood as the cognitive way of managing the intake of emotionally arousing information. (Thompson, 1991). Cognitions or cognitive processes help us manage or regulate emotions or feelings, to keep control over them and/or not get overwhelmed by them, during or after experience of threatening or stressful events. Research regarding coping tries to explain why some individuals react better than others when encountering threats, negative events, and stress in their lives. There are other concepts such as culture, personal experiences, or personality, that can help explain these individual differences, but coping is one process that lends itself to cognitive-behavioral intervention. Coping must be approached not only as an explanatory concept regarding variability in response to stress, threats or negative events, but also as a portal for intervention (Folkman & Moskowitz, 2004).

OBJECTIVE

The objective of this study was the adaptation and standardization of CERQ in the Romanian population.

DESCRIPTION OF CERQ

Though Lazarus's coping approach is one generally accepted and most frequently used, many instruments are based on it, Garnefsky, Kraaij and Spinhoven (2002b) try to explain that problem-focused and emotion focused coping is not the only dimension by which coping strategies can be classified. They discuss the cognitive as well as the behavioral dimensions of coping (Garnefsky et al., 2002a; Garnefsky, Kraaij, & Spinhoven, 2001; Holahan, Moss & Schaeffer, 1996). An example of cognitive problem-oriented coping is „making plans”; an example of behavioral problem-oriented coping is „taking immediate action”. Garnefsky, Kraaij & Spinhoven (2002b) discuss how most of the existing coping instruments are a mixture of cognitive and behavioral coping strategies. For example „making plans” (thinking about what you will do) and „taking action” (actually acting) are categorized under the same dimension, even though they refer to different processes that are used at different moments in time. Additionally, „making plans” does not always mean that they will also be carried out. You might *think* about „making plans” but not actually *act* on them. Until now it has not been possible to measure cognitive coping strategies separately from behavioral coping strategies.

Although in the past few decades the relationship between various coping strategies and psychopathology has clearly been established (for reviews see Folkman & Markowitz, 2004; Garnefsky et al., 2002a; Garnefsky et al., 2001; Endler & Parker, 1990), not much is known about certain influences that could be specifically attributed to cognitive aspects of coping (Garnefsky et al., 2002b). That is one reason why Garnefsky et al. (2002b) considered it important to have instruments that measure explicitly cognitive aspects of coping. Although considerable attention has been given to cognitive processes as regulating mechanisms for certain developmental processes, there is not much known about the degree to which cognitive coping strategies regulate emotions and how it influences the course of emotional processing after experiencing negative life events (Garnefsky et al., 2002a). Garnefsky et al. (2002b) have developed CERQ in order to fill this gap. CERQ is an instrument that gives access only to the cognitive aspects of coping, so that we can see the difference between „thinking” about something and „actually acting” and its influence on facing a negative life event. The CERQ therefore measures cognitive coping strategies exclusively, separate from the behavioral coping strategies (Garnefsky et al., 2002b).

The CERQ is a self-report questionnaire measuring coping/emotion regulation strategies of adults and adolescents aged 13 years and more. So with CERQ we can find out what people think after they have experienced a negative life event. Cognitive coping strategies refer to rather stable styles of dealing with negative life events. The way we deal with a negative life event, our coping style is stable, but not as stable as personality traits (Garnefsky et al., 2002b). Some call the coping strategies “personality in action under stress”. This means that there is a

relationship between our personality traits and the way we choose to cope with negative life events. According to some studies, personality traits mediate the relationship between coping strategies and the “result” of the coping process (Cohen & Lazarus, 1973 cited in Folkman Lazarus, Gruen, & DeLongis, 1986).

Our coping strategies are also sensitive to context, to the stressor, that is why in some situations we can use a certain coping strategy and a completely different one in other situations. It may also be assumed that potential cognitive coping strategies can be influenced, changed, learned or unlearned for example through psychotherapy, intervention programs or one’s own experience (Garnefsky et al., 2002b). Either way, knowing what cognitive coping strategies one uses when dealing with a negative life event is a portal for therapeutic intervention. You can see what the „resources” that the client brings into the therapy are and the „cognitive material” that you work with. Measuring one’s cognitive coping strategies can unfold vulnerabilities or strengths in dealing with negative or stressful life events.

CERQ has 36 items, each referring exclusively to what a person thinks and not what a person actually does when facing a negative or stressful life event. The items are divided proportionally over nine subscales, each scale has 4 items (Garnefsky et al., 2002b). Thus, the questionnaire distinguishes among nine different cognitive coping strategies (Garnefsky et al., 2002b): self-blame, acceptance, rumination, positive refocusing, refocus on planning, positive reappraisal, putting into perspective, catastrophizing, other-blame (Garnefsky et al., 2002b). Clinical psychological literature associates more often some of these coping strategies with pathology (Garnefsky et al., 2002b).

METHOD

Participants

Three samples of participants were included in this study: adolescents, adults, and psychiatric patients.

The adolescent sample comprised 368 adolescents aged 13 to 18 years ($M = 15.40$, $SD = 1.57$), 171 (46.50%) boys and 197 girls (53.50%).

The adult sample comprised 1071 adults from the general population, 18 to 65 years ($M = 39$, $SD = 10$), 372 (35%) men and 699 (65%) women.

The psychiatric patients sample comprised 182 patients 18 to 67 years ($M = 44.22$, $SD = 13.33$), 97 (58%) men and 85 (42%) women.

The participants come from different counties across Romania (Cluj, Oradea, Satu-Mare, Baia-Mare, Hunedoara, Ialomița, Galați). Data was collected in high schools, universities, companies, mental hospitals and medical clinics, other work places. The scales were administered individually or in group, depending on

the situation. All participants volunteered for the study and gave their informed consent before filling in the scales.

Procedure

The study consisted of four phases: 1) scale forward and back translation; 2) a pilot study for verification of translated items; 3) determination of validity and reliability.

The first phase of forward and backward translation was completed in one week. The translation aimed at the conceptual equivalent of a word or phrase, not a word-for-word translation (not a literal translation). Technical and highly scientific terms and expressions were avoided. Considering that the questionnaire is also for adolescent population, that language was adequate for this age group.

The first Romanian translation was subject to discussions, questioning, and suggesting alternatives for certain words or expressions. The expert panel included the original translator and four other specialists in psychology. After all the discussions they agreed on an initial version of CERQ in the Romanian language.

This completed translated version was back-translated by another independent translator who had no knowledge of the original version of CERQ. As in the initial translation, emphasis in the back-translation was on conceptual and cultural equivalence.

A preliminary version was obtained after additional discussions with the experts.

In the second phase a pilot study was initiated to verify the accuracy of translated items. The preliminary version of CERQ was administered to 30 persons in order to test the instructions, item comprehension, and the ease of administration.

Few changes were made in relation to the given version, e.g., reformulation of the instructions in order to improve their clarity, change a few words to better conform to the spoken language. After summarizing and analyzing all the problems found and all the modifications were implemented, the final Romanian version of CERQ was ready for use.

Measures

The validation study included measures of coping strategies, personality traits, anxiety, and depressive symptoms, which are described in detail below.

Coping strategies

Other coping strategies were measured using the Strategic Approach to Coping Scale (SACS), a questionnaire developed by Hobfoll, Dunahoo, Monnier, Hulsizer and Johnson (1998). SACS is a 52-item questionnaire presented on a 5-point Likert scale which measures mostly behavioral coping strategies. It is based on the multi-

axial model of coping (Hobfoll, Dunahoo, Monnier, Hulsizer, & Johnson, 1998; Hobfoll, Dunahoo, Ben-Porath, & Monnier, 1994). SACS measures coping strategies considering three dimensions: active/prosocial, active/antisocial, active/passive. It is a questionnaire that measure coping strategies emphasizing the social context and the environment where a person lives. The authors approached the multi-axial model of coping trying to explain coping strategies in a more complex/complete context, considering at the same time individualistic and inter-social aspects of coping.

Personality traits

Personality traits such as extraversion, agreeableness, conscientiousness, emotional stability, autonomy, and social desirability were measured using the Five-Factor Personality Questionnaire (CP5F). CP5F was developed by Monica Albu (2008) and it evaluates the five factors of the Big Five Model. It can be used in personality diagnosis, educational and clinical context and health psychology. It has 130 items, some of them negatively keyed, and grouped in 6 subscales: extraversion, agreeableness, conscientiousness, emotional stability, autonomy, social desirability.

Anxiety and depression symptoms

Anxiety and depression symptoms were measured using the Depression, Anxiety and Stress Scale - DASS (Lovibond & Lovibond, 1995). The Depression, Anxiety, Stress Scale corresponds with the tripartite model of anxiety and depression (Clark & Watson, 1991). This model suggests that anxiety and depression have both unique and shared features. DASS was designed to measure the core symptoms of anxiety and depression. The Romanian version of DASS has 21 items, grouped on three subscales: anxiety, depression, and stress.

Results

In order to define the dimensional structure of CERQ, a Principal Component Analysis with Varimax-rotation on item level was performed for the groups of adolescents and adults. The factor loadings matrix for the two groups is presented in Table 1 and Table 2. These tables show all factor loadings greater than .40 for the two groups (values below .40 are put between brackets).

For the adolescent group the curves of the plotted Eigen values showed a ten factor solution. All the factors had an Eigen value greater than 1 ($\lambda > 1$). The values of the communalities ranged from .41 to .80. In the population of adolescents the ten factors together explained in all 60.79% of the variance. As Table 1 shows, only two items (7 and 8) loaded on the 10th factor, and item 7 loaded more strongly on the factor that belonged to *Putting into perspective*. As Table 1 shows, the

factors found were consistent with the structure proposed by Garnefski et al. (2002b), especially for *Self-blame*, *Positive refocusing*, *Putting into perspective*, and *Other-blame*. Almost all items included in one and the same dimension *on a theoretical basis*, turned out to actually load on one and the same dimension *on an empirical basis*. Some deviations from the proposed structure were found, though. There were scales that had at least one item that loaded on a different factor than the one that belonged to. In other cases (*Acceptance*, *Refocus on planning*, and *Catastrophizing*) the deviations from the accepted structure were very small (two deviant items). Big deviations were observed in case of *Rumination* (two items loaded stronger on *Catastrophizing*) and *Positive reappraisal*. In case of *Positive reappraisal* two items loaded stronger on *Refocus on planning*. This overlap between *Positive reappraisal* and *Refocus on planning* was also identified by Garnefski et al. (2002b), in both adolescent and the adult populations.

For adults, the curves of the plotted Eigen values showed an eight factors solution, the 9th factor having an Eigen value smaller than 1 ($\lambda = .92$). The eight factor solution explained 60.29%, while the nine factor solution explained 62.85% of the total variance. As Table 2 shows, the factor structure in this group proved to be roughly similar to the original one obtained by Garnefski et al. (2002b). There are a few exceptions though. Two items (30, 31) from *Rumination* and *Positive refocusing* scales had factor loadings which all turned out to exceed .40 on a different factor than the one that they belonged to. In both of these situations the loadings were equal to or smaller than the loadings on factors theoretically adequate. Item 19, which theoretically should have loaded on the dimension made up by the items belonging to *Self-blame*, appeared to load much stronger on the dimension belonging to *Rumination*, *Refocus on planning*, and *Positive reappraisal*. A careful inspection of the internal consistency will clarify to what extent keeping this item on *Self-blame* scale is justified. Items of *Positive Reappraisal*, and *Refocus on planning* ended up on the same dimension. The situation is identical with the one found by Garnefski et al. (2002b) in the process of validation of original version of the scale. Here again, it is true that a careful inspection of the internal consistency of the two scales is important.

As this analysis shows, the structure corresponds largely with the theoretical structure proposed by Garnefski et al. (2002b), especially in case of the adult population. For adolescents, the overlap between the Romanian version and the original version of the scales structure is not so strong, that is why other studies are needed, eventually with a bigger number of subjects.

Table 1.
Factor loadings PCA after Varimax rotation, adolescents group

	Components									
	1	2	3	4	5	6	7	8	9	10
Self-blame										
CERQ1						.59				
CERQ10						.80				
CERQ19						.42				
CERQ28						.62				
Acceptance										
CERQ2							.66			
CERQ11							.50			
CERQ20			.56				(.49)			
CERQ29							.67			
Rumination										
CERQ3									.80	
CERQ12									.55	
CERQ21			.49						(.37)	
CERQ30			.60						(.18)	
Positive refocusing										
CERQ4		.68								
CERQ13		.80								
CERQ22		.78								
CERQ31		.72								
Refocus on planning										
CERQ5					(.43)			.48		
CERQ14					.49					
CERQ23					.71					
CERQ32					.68			(.41)		
Positive reappraisal										
CERQ6								.71		
CERQ15								.54		
CERQ24					.57			(.13)		
CERQ33				(.50)	.48			(.08)		
Putting into perspective										
CERQ7				.57						(.51)
CERQ16				.69						
CERQ25				.59						
CERQ34				.72						
Catastrophizing										
CERQ8	.40		(.12)							.52
CERQ17			.64							
CERQ26	.48		(.35)							
CERQ35			.74							
Other-blame										
CERQ9	.75									
CERQ18	.71									
CERQ27	.68									
CERQ36	.67									

Table 2.
Factor loadings PCA after Varimax rotation, Adults group

	Components							
	1	2	3	4	5	6	7	8
Self-blame								
CERQ1								.75
CERQ10								.74
CERQ19	.45					.42		(.28)
CERQ28								.72
Acceptance								
CERQ2							.66	
CERQ11							.70	
CERQ20							.66	
CERQ29							.63	
Rumination								
CERQ3						.75		
CERQ12						.71		
CERQ21						.64		
CERQ30			(.45)			.45		
Positive refocusing								
CERQ4		.81						
CERQ13		.79						
CERQ22		.79						
CERQ31	(.43)	.62						
Refocus on planning								
CERQ5	.56							
CERQ14	.65							
CERQ23	.73							
CERQ32	.80							
Positive reappraisal								
CERQ6	.51							
CERQ15	.56							
CERQ24	.57							
CERQ33	.67							
Putting into perspective								
CERQ7				.69				
CERQ16				.69				
CERQ25				.67				
CERQ34				.64				
Catastrophizing								
CERQ8			.52					
CERQ17			.70					
CERQ26			.70					
CERQ35			.73					
Other-blame								
CERQ9					.75			
CERQ18					.80			
CERQ27					.58			
CERQ36					.72			

Construct validity

Discriminative properties of a test are very important in order to prove the test validity. Considering the literature in the field, it is assumed that the mean score should be higher in the psychiatric population than in the non-clinical population especially on certain CERQ scales, such as: *Self-blame*, *Catastrophizing*, *Rumination* (Garnefski et al., 2002b). Studies have shown that personality traits influence rather than determine coping strategies (Cohen & Lazarus, 1973 cited in Folkman et al., 1986).

The assessment of construct validity of CERQ was performed: a) by analyzing the correlations between CERQ and CP5F; b) by identifying the effectiveness of CERQ scales in differentiating between clinical and non-clinical population; c) by analyzing the differences between CERQ and another measure of coping strategies, performing an factorial analysis for CERQ and SACS; and d) analyzing the relation between coping strategies and measures of anxiety and depression, measured here by DASS.

Correlations between CERQ and personality traits

The relationship between coping strategies and certain personality traits was analyzed. Personality traits were measured with CP5F (Albu, 2008). CERQ and CP5F were applied on a non-clinical sample, 34 persons aged between 20 and 50 years ($M = 30.5$, $SD = 9.86$), 20 women and 14 men.

Not too high correlations are expected between the five personality factors and cognitive coping strategies, because cognitive coping strategies measure something else than personality traits. As expected, CERQ subscales correlated with personality factors measured by CP5F. Results are presented in Table 3. The relationship between coping strategies and personality traits is as expected. We can see, for example, that *Emotional Stability*, which is often associated with functional coping strategies, correlates with *Positive Refocusing*, *Refocus on planning*, and *Positive Reappraisal*. *Extraversion* correlates with *Positive Refocusing*, but with *Rumination* also. *Autonomy* was also associated with more adaptive coping strategies such as *Positive Refocusing*, *Refocus on planning* and the correlation is negative with *Catastrophizing*. There is a tendency of *Consciousness* to correlate more with dysfunctional coping strategies, such as *Self-blame*, *Rumination*, but at the same time it correlates with *Acceptance* and *Refocus on planning*, coping strategies considered rather adaptive/functional.

The relationship between coping strategies and personality traits is not one of cause and effect. Personality traits might have an influence on the way a person deals with a negative life event. When we talk about the personality traits and coping strategies relationship we talk in terms of probability. So there is a probability for a person who has certain personality traits to use certain coping

strategies. For example there is a probability for someone who scores high in *Emotional Stability* to use more adaptive coping strategies when facing a negative life event, but this is not a guarantee that he might not use dysfunctional coping strategies when dealing with another negative life event. Results of this analysis confirm the relationship between the CERQ scales and the personality traits (see also Folkman & Moskowitz, 2004).

Table 3.
Correlations between CERQ scales and CP5F

CERQ subscales	Personality scales (CP5F)				
	Ext	Em. St.	Consc.	Amab	Auton
Self-blame	.40	-.35*	.43*	.32*	-.08
Acceptance	.40	.37*	.54*	.51*	.18
Rumination	.54*	.33*	.55*	.36*	.11
Positive refocusing	.58*	.76**	-.24	.34*	.59*
Refocus on planning	.22	.63*	.33*	-.04	.50*
Positive reappraisal	.23	.46*	.02	.44*	.36*
Putting into perspective	-.03	.54*	.26	.34*	-.07
Catastrophizing	.29	-.41*	-.12	-.19	-.41*
Other-blame	.12	-.36*	.03	-.05	-.03

* $p < 0,05$; ** $p < 0,01$

Note: Ext = Extraversion; Consc = Conscientiousness; Agree = Agreeableness; Em. St. = Emotional stability; Auton = Autonomy.

Comparisons between clinical and non-clinical populations

According to the literature in the field, certain coping strategies are more often associated with psychopathology than others. Coping is also strongly associated with emotion regulation, especially distress (Folkmann & Moskowitz, 2004). As we

mentioned before, coping is a process that unfolds in an emotional environment (Folkmann & Moskowitz, 2004), and especially when we deal with a negative life event. To see if there are differences between the coping strategies that clinical and non-clinical population use, *independent t tests* were performed. In the study participated a clinical and a non-clinical sample. The clinical and non-clinical samples are described above in the *Participants* section. People in the clinical sample were diagnosed with Anxiety Disorders, Depression, Personality Disorders, Hypochondria, Alcohol Abuse, also mixed anxiety and depression, Bipolar Disorder.

As expected, *independent t test* results showed that there were significant differences between the two groups. Table 4 presents the mean differences between the two groups on each CERQ subscale. The most significant differences appear in the case of those coping strategies considered dysfunctional (*Catastrophizing, Self-blame, Rumination*). Scores are higher in the non-clinical sample on those coping strategies considered more adaptive (*Positive refocus, Positive Reevaluation, Putting into perspective*), compared with the clinical sample. For *Acceptance* the mean difference between the clinical and non-clinical sample is not that strong. A possible explanation for this might be that *Acceptance* as a coping strategy is not very often associated either with positive, or with negative mental health outcomes. Sometimes *Acceptance* can be functional, sometimes dysfunctional. The fact that we accept that something bad happened to us doesn't mean that our negative emotions are less intense. Even if we accept a negative life event we might feel sad, or angry, or anxious. Sometimes the fact that we accept what happened frees us to move on. These results show that there is a difference between clinical and non-clinical populations and we can see from the data that certain coping strategies are more associated with pathology, but we can't say anything about the type of this relationship. We don't know if the pathology leads to a frequent use of certain coping strategies or if certain coping strategies (for example *Catastrophizing*) leads to pathology.

Table 4.
Mean differences between clinical and non-clinical population

Scale	Sample	N	Mean	Standard deviation	T	Sig. (2-tailed)	Mean difference
Self-blame	Clinic	178	11.54	3.45	6.89	.000	1.88
	Non-Clinic	1059	9.66	2.82			
Acceptance	Clinic	170	12.44	2.88	3.22	.001	.78
	Non-Clinic	1076	11.66	3.28			
Rumination	Clinic	176	13.59	3.90	8.25	.000	2.57
	Non-Clinic	1068	11.02	3.36			
Positive refocusing	Clinic	179	9.87	4.07	4.22	.000	-1.31
	Non-Clinic	1084	11.19	3.82			
Refocus on planning	Clinic	173	12.82	3.54	5.38	.000	-1.52
	Non-Clinic	1080	14.33	3.42			
Positive reappraisal	Clinic	175	9.98	3.95	12.70	.000	-4.04
	Non-Clinic	1083	14.02	3.59			
Putting into perspective	Clinic	179	10.45	3.60	7.64	.000	-2.39
	Non-Clinic	1083	12.83	3.91			
Catastrophizing	Clinic	176	13.47	3.85	17.49	.000	5.37
	Non-Clinic	1084	8.10	3.27			
Other-blame	Clinic	182	10.86	4.23	9.24	.000	3.00
	Non-Clinic	1080	7.86	2.75			

CERQ and SACS

In order to show that CERQ measures a certain coping dimension, the relationship between CERQ and SACS (another coping questionnaire) was analyzed. CERQ measures the cognitive dimension of coping while SACS measures coping strategies considering more social and behavioral aspects of coping. A factor analysis was performed in order to show that the two questionnaires, although they measure the same construct (coping), each measures different aspects of it.

Factor analysis was performed on the scales of the two coping questionnaires, using Principal Components Method and Varimax rotation. The sample consisted of 105 persons (part of the Adults sample), 43 men and 60 women, 2 persons didn't mark their gender, age 20 to 67 years ($M = 38.21$; $SD = 12.50$). The factor analysis extracted 6 factors that explain 69.98% from the variance ($F1$: 13.78%; $F2$:12.63%; $F3$:12.35%; $F4$: 10.83%; $F5$:10.80%; $F6$:8.58%). Results are presented in Table 5.

Results show that items of CERQ load on totally different factors than items of SACS, this means that the two questionnaires measure different things, even if in this case we talk about different aspects of the same construct.

Table 5.
Factor analysis for the CERQ and SACS scales

Scale	Factor					
	F1	F2	F3	F4	F5	F6
<i>Assertive action(SACS)</i>	.179	.120	.079	.871	.011	.002
<i>Social joining(SACS)</i>	.061	.851	.109	.004	.063	.001
<i>Seeking social support(SACS)</i>	.068	.837	-.012	.196	-.033	.005
<i>Cautious action(SACS)</i>	-.029	.754	.093	-.234	.064	-.054
<i>Instinctive action(SACS)</i>	.244	.087	.520	-.437	.028	-.347
<i>Avoidance(SACS)</i>	-.005	.111	.106	-.851	-.060	-.139
<i>Indirect action(SACS)</i>	-.018	.156	.750	-.115	.010	-.015
<i>Antisocial action(SACS)</i>	-.002	-.192	.846	-.040	.072	.057
<i>Aggressive action(SACS)</i>	.129	.268	.718	.233	.062	-.085
<i>Self-blame(CERQ)</i>	.017	-.025	.136	.113	.841	.014
<i>Acceptance (CERQ)</i>	.228	.223	-.141	.051	.659	.063
<i>Rumination(CERQ)</i>	.176	-.042	.102	-.099	.746	.110
<i>Positive refocusing (CERQ)</i>	.662	-.034	.194	.140	-.087	.159
<i>Refocus on planning(CERQ)</i>	.818	-.050	-.056	-.008	.174	.009
<i>Positive reappraisal(CERQ)</i>	.851	.072	-.049	.006	.128	-.117
<i>Putting into perspective (CERQ)</i>	.662	.137	.088	.038	.225	.086
<i>Catastrophizing (CERQ)</i>	.087	-.172	.087	.261	.344	.710
<i>Other- blame(CERQ)</i>	.061	.079	-.122	-.004	-.009	.908

CERQ and DASS-21

DASS (Lovibon & Lovibond, 1995) and CERQ were applied together in order to analyze the relationship between cognitive coping strategies and anxiety and depression. In the study participated 1030 adults from the general population, 361 women and 669 men, age 19 to 65 years ($M = 38.84$, $SD = 10.10$). It is expected that less functional coping strategies correlate stronger with depression and anxiety scales while those coping strategies considered more functional will have low correlations with anxiety and depression scales (Garnefski, Teerds, Kraaij, Legerstee, & Van den Kommer, 2003; Kraaij, Garnefski, & van Gerwen, 2003). Correlations between DASS-21 and CERQ scales are presented in Table 6. As it was expected, strong relationships appear to exist between the *Catastrophizing*, *Other-blame*, *Self-blame*, and *Rumination* and DASS-21 scales. Those coping strategies considered more functional (Positive reappraisal, Putting into perspective, Positive refocusing) correlate less with DASS-21 scales. Results confirm the expectancies that less functional coping strategies correlate stronger with depression, anxiety, and stress, measured here by DASS-21 (Perțe & Albu, 2011).

Table 6.
Correlations between DASS-21 and CERQ subscales

CERQ Subscales	DASS-21 Anxiety	DASS-21 Stress	DASS-21 Depression
Self-blame	.18**	.19**	.20**
Acceptance	.12**	.10**	.13**
Rumination	.17**	.20**	.21**
Positive refocusing	.03	.00	.04
Refocus on planning	-.04	-.02	-.02
Positive reappraisal	-.02	-.03	-.03
Putting into perspective	.03	.04	.03
Catastrophizing	.33**	.28**	.33**
Other-blame	.26**	.23**	.29**

*: $p < 0,05$; **: $p < 0,01$.

Reliability

Alpha Cronbach and test-retest analyses were performed in order to test CERQ's reliability.

Internal consistency: alpha Cronbach

To assess the internal consistency of the nine CERQ scales alpha coefficients were calculated in all research groups, the outcome of which is presented in Table 7.

Generally the alpha coefficients values range from .63 to .84, with the exception of *Acceptance* scale ($\alpha = .59$) in the Adolescent group and the Psychiatric patients group ($\alpha = .48$). In the case of adults, with the exception of *Self-blame*, which has an acceptable internal consistency ($\alpha = .69$), for all the other scales alpha coefficients values range from .71 to .83. Same thing can be told about the Psychiatric patients where the alpha coefficients range from .73 to .84, with exception of *Acceptance* ($\alpha = .48$) and *Rumination* ($\alpha = .65$). Considering the number of items on each scale, even small values like .59 for *Acceptance* in the Adolescents sample or .48 in the Psychiatric patients can be considered acceptable.

Table 7.
Alpha Cronbach coefficients of the CERQ subscales

Subscales	Adolescents N (368)	Adults N(1071)	Psychiatric patients N (182)
Self-blame	.66	.69	.81
Acceptance	.59	.71	.48
Rumination	.63	.76	.65
Positive refocusing	.79	.83	.84
Refocus on planning	.69	.80	.73
Positive reappraisal	.69	.80	.81
Putting into perspective	.71	.75	.73
Catastrophizing	.64	.76	.78
Other-blame	.72	.75	.75

Stability (test-retest reliability)

The CERQ was administered twice to a group of 50 adults from the general population. The data was used to compute test-retest correlations. There was a one month interval between the two measurements. The results must be interpreted considering the short period of time between the two measurements. Table 8 presents the test-retest correlations, means and standard deviations and results of *paired t test*. The test-retest correlations range from $r = .42$ ($p < .05$; *Catastrophizing*) to $r = .64$ ($p < .001$; *Positive reappraisal*). These values suggest that we are talking about relatively stable styles of coping, considering the short period of time. Coping strategies are not as stable as personality traits, so other factors like the stressor, the context, personality traits, the control that one has in a given situation can influence the coping that we adopt when we deal with a negative life event (Terry & Hynes, 1998).

Results show that the majority of the CERQ scales measure rather stable coping styles. For three of the CERQ scales the correlations were not significant (*Acceptance* ($r = .34$; $p = .10$), *Positive refocusing* ($r = .18$; *ns*), *Other-blame* ($r = .28$; *ns*). On the other hand, results from paired *t test*, which test whether the mean individual difference scores of the first and second measurements deviate significantly from zero, showed that mean differences are not significant between pre- and post-measurement for none of the CERQ scales. This explains that there

are reasons to consider that CERQ scales are stable in time. In order to get a more accurate perspective on CERQ scales stability in time, data must be collected from a large sample of persons and in a longer period of time (6 months for example).

Table 8.

Test – retest coefficients of CERQ scales after one month (adults from the general population, age 21-30 years).

Subscales	r_{1-2}	N	T1 M(SD)	T2 M(SD)	Test t (paired)
Self-blame	.62***	50	11.32 (2.12)	10.88 (2.86)	.97
Acceptance	.34	50	13.80 (1.98)	13.40 (3.38)	.61
Rumination	.47*	50	14.00 (3.89)	13.64 (4.74)	.40
Positive refocusing	.18	50	11.12 (3.32)	9.72 (3.06)	1.71
Refocus on planning	.56**	50	16.52 (2.47)	15.64 (3.20)	1.60
Positive reappraisal	.64***	50	15.24 (3.95)	14.20 (4.39)	1.45
Putting into perspective	.53**	50	14.32 (3.72)	13.00 (4.46)	1.65
Catastrophizing	.42*	50	7.92 (2.25)	7.60 (2.83)	.58
Other-blame	.28	50	8.44 (2.04)	8.32 (2.46)	.22

* $p < 0,05$; ** $p < 0,01$; *** $p < 0,001$

Limits

Coping questionnaires are helpful considering that people can give information about thoughts and behaviors they adopt when dealing with a negative life event. Nevertheless, the inventory approach has many limits that our study also confronts.

One of the most prominent criticisms concerns the problem of retrospective report and the accuracy of recall about specific thoughts and behaviors that were used one week, or month or even more time earlier (Stone & Neale, 1984). Momentary and retrospective accounts yield different information about coping (Folkman & Moskowitz, 2004). Our coping strategies might change during the same stressful/negative life event, before and after, so the retrospective recall has its shortcomings. Life experiences can also have a big influence on our coping

strategies. In short, the limits that our study deals with are: variations in the recall period and unreliability of recall.

CONCLUSIONS

From the various Principal Component Analyses there clearly emerge comparable pictures between the Romanian version and the original version of CERQ scales. In all cases the dimensions explain over 60% of the variance. In most cases the dimensions are in full accord with the scales established on a theoretical basis. The only consistent exception is the overlap between the items belonging to the *Refocus on Planning* and *Positive Reappraisal* scales. In most cases these items ended up on one and the same dimension. This is probably due to the rather strong correlation between these two scales (.50 in the adolescent population to .70 in the adult population). On a theoretical basis, it is important to keep distinguishing these two subscales clearly as two different concepts. While the concept of *Refocus on Planning* clearly focuses on thinking about what steps to take in order to cope with the event (action-oriented), the concept of *Positive Reappraisal* focuses on attributing a positive meaning to the event in terms of personal growth (emotion-oriented). Still, the Principal Components Analyses and the correlation analyses make it clear that the two concepts are closely linked. Therefore, this is certainly important to take into account when interpreting the scores. Considering these results, we can say that CERQ has proven to be a reliable and valid tool for assessing cognitive coping strategies.

The present study focused on the adaptation on the Romanian population of CERQ scales. We analyzed different relationships between coping strategies and other constructs, coping strategies, and pathological conditions.

Results of this study confirm the relationship between coping strategies and personality traits. Certain personality traits predispose us to use certain coping strategies. We don't know yet if it is a cause and effect relationship. Watson, David and Suls (1999, p. 119) consider that coping strategies reflect broader and more basic dispositional tendencies within the individual and that there is a relationship between personality traits and the coping strategy an individual chooses when facing a negative life event. Future studies will have to find more about this.

Our results also confirm the fact that use of certain coping strategies (for example *Cathastrophizing*) is associated with psychopathology and use of other coping strategies (for example *Putting into perspective*) is associated with mental health. The relationship is not one of cause and effect so we don't know yet if use of certain coping strategies leads to pathology or if pathology leads to use of those coping strategies considered dysfunctional.

These findings correspond with the expectations that hold for the concept of cognitive coping strategies and support the assumption that although cognitive

coping strategies refer to personal coping styles, it should potentially be possible to influence, change, learn, and unlearn them (Garnefskiet *al.*, 2002b). This is an important point for mental health intervention.

We can say that coping strategies like *Catastrophizing* and *Rumination*, for example, are more often associated with poor mental health while coping strategies like *Putting into perspective* and *Positive reappraisal*, for example, are more often associated with mental health. This should raise a question for clinicians especially. Even if we don't know anything yet about the relationship between coping strategies and mental health, the fact that results show that there is an association between certain coping strategies and mental health should be a step in intervention. Working on functional cognitive coping strategies in psychotherapy can be a part of the cognitive restructuring. The fact that we know what kind of coping strategies are associated with mental health can make a difference.

ACKNOWLEDGEMENTS

We are thankful to all the participants in the study, all the people involved in data gathering and those who organized the data. We also acknowledge the assistance of the schools, university departments, hospitals and other work places which granted the permission to administer the CERQ.

REFERENCES

- Albu, M. (2008). Suprafactorii modelului Big Five (CP5F) [The big factors of the Big Five Model]. *CAS +- Cognitrom Assessment System*, -Vol. 3. Personalitate și interese [Personality and interests]. Cluj-Napoca: ASCR.
- Cicchetti, D., Ackerman, B. P., & Izard, C. E. (1995). Emotions and emotion regulation in developmental psychopathology. *Development and Psychopathology*, 7, 1–10.
- Clark, L. A., & Watson, D. (1991). Tripartite model of anxiety and depression: Psychometric evidence and taxonomic implications. *Journal of Abnormal Psychology*, 100, 316–336.
- Compas, B. E., Orosan, P. G., & Grant, K. E. (1993). Adolescent stress and coping: Implications for psychopathology during adolescence. *Journal of Adolescence*, 16, 331–349.
- Endler, N. S., & Parker, J. D. A. (1990). Multidimensional assessment of coping: A critical evaluation. *Journal of Personality and Social Psychology*, 58, 844–854.
- Folkman, S., Lazarus, R. S., Gruen, R., & DeLongis, A. (1986). Appraisal, coping, health status, and psychological symptoms. *Journal of Personality and Social Psychology*, 50, 571–579.
- Folkman, S., & Moskowitz, T. (2004). Coping: Pitfalls and Promise. *Annual Review of Psychology*, 55, 745–774.

- Garnefski, N., Kraaij, V., & Spinhoven, Ph. (2001). Negative life events, cognitive emotion regulation and emotional problems. *Personality and Individual Differences*, 30, 1311-1327.
- Garnefski, N., Kraaij, V., & Spinhoven, Ph. (2002b). *Manual for the use of the Cognitive Emotion Regulation Questionnaire*. Leiderdorp, the Netherlands: DATEC.
- Garnefski, N., Teerds, J., Kraaij, V., Legerstee, J., & Van den Kommer, T. (2003). Cognitive emotion regulation strategies and depressive symptoms: Differences between males and females. *Personality and Individual Differences*, 25, 603-611.
- Garnefski, N., van den Kommer, T., Kraaij, V., Teerds, J., Legerstee, J., & Onstein, E. (2002a). The relationship between cognitive emotion regulation strategies and emotional problems. *European Journal of Personality*, 16, 403-420.
- Hobfoll, S. E., Dunahoo, C. L., Ben-Porath, Y., & Monnier, J. (1994). Gender and coping: The dual-axis model of coping. *American Journal of Community Psychology*, 22, 49-82.
- Hobfoll, S. E., Dunahoo, C. L., Monnier, J., Hulsizer, M. R., & Johnson, R. (1998). There's more than rugged individualism in coping. Part 1: Even the Lone Ranger had Tonto. *Anxiety, Stress, Coping: An International Journal*, 11(2), 137-65.
- Holahan, C. J., Moos, R. H., & Schaeffer, J. A. (1996). Coping, stress resistance and growth: Conceptualizing adaptive functioning. In M. Zeidner & N.S. Endler (Eds.), *Handbook of coping* (pp. 24-43). New York: John Wiley & Sons.
- Kraaij, V., Garnefski, N., & van Gerwen, L. (2003). Cognitive coping and anxiety symptoms among people who seek help for fear of flying. *Aviation, Space, and Environmental Medicine*, 74, 273-277.
- Lazarus, R. S. & Folkman, S. (1984). *Stress Appraisal and Coping*. New York: Springer.
- Lovibond, S. H., & Lovibond, P. F. (1993). *Manual for the Depression Anxiety Stress Scales (DASS)*. Sidney, Australia: University of New South Wales.
- Monat, A., & Lazarus, R. S. (1991). *Stress and coping: An anthology*. New York: Columbia University Press.
- Perțe, A., & Albu, M. (2011). *Adaptarea și standardizarea pe populația din România. DASS-Manual pentru Scalele depresie, anxietate și stres* [The adaptation and standardisation on roumanian population. DASS- Manual for depression, anxiety and stress scales]. Cluj-Napoca: ASCR.
- Stone, A. A., & Neale, J. M. (1984). New measure of daily coping: Development and preliminary results. *Journal of Personality and Social Psychology*, 46(4), 892-906.
- Terry, D. J., & Hynes, G. J. (1998). Adjustment to low-control situation reexamining the role of coping responses. *Journal Of Personality and Social Psychology*, 74(4), 1078-1092.
- Thoits, P. A. (1995). Stress, coping, and social support processes: Where are we? What next? [Extra Issue]. *Journal of Health and Social Behavior*, 35, 53-79.
- Thompson, R.A. (1991). Emotional regulation and emotional development. *Educational Psychology Review*, 3(4), 269-307.
- Watson, D., David, J. P., & Suls, J. (1999). Personality, Affectivity, and Coping. In C.R. Snyder (Ed.), *Coping: The psychology of what works* (pp. 119-138). New York: Oxford University Press.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.